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LINEAR ALGEBRA

2.1 Matrix

An array of elements in horizontal lines (Rows) and Vertical Lines (Columns) is called a Matrix.

Example: $A = \begin{bmatrix} i & n & d & i & a \\ j & a & p & a & n \end{bmatrix}$

2.1.1 Size of Matrix

If a matrix has 'm' rows and 'n' columns, then we say that the size of the matrix is $m \times n$ (read as m by n)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \dots a_{1n} \\ a_{21} & a_{22} & a_{23} \dots a_{2n} \\ \vdots & \vdots & \ddots \\ a_{m1} & a_{m2} & a_{m3} \dots a_{mn} \end{bmatrix}; A = [a_{ij}]_{m \times n} \text{ such that } 1 \leq i \leq m, 1 \leq j \leq n \text{ and } a_{ij} = f(i, j)$$

2.1.2 Addition of Matrices

- (i) Two matrices $A = [a_{ij}]_{m \times n}$ & $B = [b_{ij}]_{p \times q}$ can be added only if $m = p$ & $n = q$.
- (ii) Matrix Addition is commutative ($A + B = B + A$)
- (iii) Matrix Addition is Associative. $A + (B + C) = (A + B) + C$
- (iv) Existence of additive identity : If O be $m \times n$ matrix each of whose elements are zero. Then, $A + O = A = O + A$ for every $m \times n$ matrix A.
- (v) Existence of additive inverse : Let $A = [a_{ij}]_{m \times n}$ then the negative of matrix A is defined as matrix $[-a_{ij}]_{m \times n}$ and is denoted by $-A$.
 $\Rightarrow$ Matrix $-A$ is additive inverse of A. Because $(-A) + A = O = A + (-A)$. Here O is null matrix of order $m \times n$.
- (vi) Cancellation laws holds good in case of addition of matrices, which is $X = -A$.
 $\Rightarrow A + X = B + X \Rightarrow A = B$
 $\Rightarrow X + A = X + B \Rightarrow A = B$
- (vii) The equation $A + X = 0$ has a unique solution in the set of all $m \times n$ matrices.

2.1.3 Multiplication of Matrices

The multiplication of two matrices $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{p \times q}$ ($\Rightarrow AB_{m \times q}$) is feasible only if $n = p$.

$$A_{m \times n} \cdot B_{p \times q} = C$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}_{3 \times 2} \quad A_{3 \times 3} \times B_{3 \times 2}$$

$$\Rightarrow \begin{bmatrix} a_{11} \cdot b_{11} + a_{12} \cdot b_{21} + a_{13} \cdot b_{31} & a_{11} b_{12} + a_{12} b_{22} + a_{13} b_{32} \\ a_{21} \cdot b_{11} + a_{22} \cdot b_{21} + a_{23} \cdot b_{31} & a_{21} b_{12} + a_{22} b_{22} + a_{23} b_{32} \\ a_{31} \cdot b_{11} + a_{32} \cdot b_{21} + a_{33} \cdot b_{31} & a_{31} b_{12} + a_{32} b_{22} + a_{33} b_{32} \end{bmatrix}_{3 \times 2}$$

2.1.4 Properties of Multiplication of Matrices

- Matrix Multiplication Need not be commutative.
- Matrix Multiplication is Associative $(A(BC)) = ((AB)C)$
- Matrix Multiplication is distributive $A(B + C) = (AB + AC)$
- The product of two Matrices $A_{m \times n}, B_{n \times q}$ (i.e. $AB_{m \times q}$) can be a zero matrix even if $A \neq 0$ & $B \neq 0$.

Example: $A = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}; B = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} \Rightarrow AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

- For the multiplication of two matrices $A_{m \times n}$ & $B_{n \times q}$
 - The No. of Multiplications required = $m \cdot n \cdot q$
 - The number of Additions required = $m(n-1)q$

2.2 Types of Matrices

- Upper triangular Matrix:** A matrix $A = [a_{ij}]; 1 \leq i, j \leq n$ is said to be an upper triangular matrix if

$$a_{ij} = 0 \quad \forall i > j$$

- Lower Triangular Matrix:** A matrix $A = [a_{ij}]_{n \times n}; 1 \leq i, j \leq n$ is said to be a lower Triangular Matrix

$$\text{if } a_{ij} = 0 \quad \forall i < j$$

- Diagonal Matrix:** A matrix $A = [a_{ij}], \forall 1 \leq i, j \leq n$ is said to be a diagonal matrix if $a_{ij} = 0 \quad \forall i \neq j$

Example: $A = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$. The diagonal Matrix is also denoted as $A = \text{diag} [d_1, d_2, d_3]$

- Scalar Matrix:** A Matrix 'A' = $[a_{ij}]; 1 \leq i, j \leq n$ is said to be a scalar Matrix if $a_{ij} = \begin{cases} K; i = j \\ 0; i \neq j \end{cases}$

If $K = 1$, then $A \rightarrow$ Identity Matrix,

If $K = 0$, then $A \rightarrow$ Null Matrix.

- Idempotent Matrix:**

A Matrix ' $A_{n \times n}$ ' is said to be an idempotent matrix if $A^2 = A$.

Example: $A = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix}$

$$\Rightarrow A \cdot A = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} = A$$



- (6) **Nilpotent Matrix:** A matrix A is said to be nilpotent of class x or index if $A^x = 0$ and $A^{x-1} \neq 0$ i.e. x is the smallest index which makes $A^x = 0$.

Example: The matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is nilpotent class 3, since $A \neq 0$ and $A^2 \neq 0$, but $A^3 = 0$.

- (7) **Orthogonal Matrix:** A matrix A is said to be orthogonal if $A \cdot A^T = I$

Example: $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

- (8) **Involutory Matrix:** A matrix A is said to be involutory if $A^2 = I$

Example: $\begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$

2.3 Transpose of a Matrix

For a given matrix $A = [a_{ij}]$; $1 \leq i \leq m, 1 \leq j \leq n$, we can say that 'B' where $B = [b_{ij}]$, $1 \leq i \leq n, 1 \leq j \leq m$ is the transpose of the Matrix 'A' if $a_{ij} = b_{ji}$

2.3.1 Properties of Transpose of a Matrix

- (i) $(A^T)^T = A$
- (ii) $(AB)^T = B^T \cdot A^T$
- (iii) $(KA)^T = KA^T$ where 'K' is a scalar.

2.4 Conjugate of a matrix

The matrix obtained by replacing each element of matrix by its complex conjugate.

2.4.1 Properties of conjugate matrix

- (a) $\overline{\overline{A}} = A$ (b) $\overline{A+B} = \overline{A} + \overline{B}$
- (c) $\overline{KA} = \overline{K} \overline{A}$ (d) $\overline{AB} = \overline{A} \overline{B}$
- (e) $\overline{\overline{A}} = A$ if A is real matrix
 $\overline{\overline{A}} = -A$ if A is purely imaginary matrix

2.5 Transposed Conjugate of a Matrix

The transpose of conjugate of a matrix is called transposed conjugate. It is represented by A^θ .

- (a) $(A^\theta)^\theta = A$
- (b) $(A+B)^\theta = A^\theta + B^\theta$
- (c) $(KA)^\theta = \overline{K} A^\theta$ (K : Complex number)
- (d) $(AB)^\theta = B^\theta A^\theta$

2.6 Trace of a Matrix

Trace is simply sum of all diagonal elements of a matrix.

2.6.1 Properties of Trace of a matrix

Let A and B be two square matrices of order η and λ is scalar then

1. $Tr(\lambda A) = \lambda Tr(A)$
2. $Tr(A+B) = Tr(A) + Tr(B)$
3. $Tr(AB) = Tr(BA)$ [If both AB and BA are defined]

2.7 Type of Real Matrix

- (a) Symmetric matrix : $(A)^T = A$
- (b) Skew symmetric matrix : $(A^T) = -A$
- (c) Orthogonal matrix : $(A^T = A^{-1}, AA^T = I)$

Note : (a) If A and B are symmetric, then $(A+B)$ and $(A-B)$ are also symmetric.

(b) For any matrix AA^T is always symmetric.

(c) For any matrix, $\left(\frac{A+A^T}{2}\right)$ is symmetric and $\left(\frac{A-A^T}{2}\right)$ is skew symmetric.

(d) For orthogonal matrices, $|A| = \pm 1$

(e) We can write any matrix A as a sum of symmetric and skew symmetric matrix $A = \frac{A+A^T}{2} + \frac{A-A^T}{2}$

2.8 Type of complex matrix

- (a) Hermitian matrix : $(A^\theta = A)$
- (b) Skew-Hermitian matrix: $A^\theta = -A$
- (c) Unitary matrix : $(A^\theta = A^{-1}, AA^\theta = I)$

Note : (a) $\frac{A+A^\theta}{2}$ is Hermitian and $\frac{A-A^\theta}{2}$ is skew Hermitian matrix.

(b) We can write any matrix as a sum of Hermitian and skew Hermitian matrix $A = \frac{A+A^\theta}{2} + \frac{A-A^\theta}{2}$

2.9 Determinant

The summation of the product of elements of a row(or) column of a matrix with their corresponding Co-factors.

$$A \cdot adj(A) = |A| \cdot I$$

Determinant can be calculated only if matrix is a square matrix.

Suppose, we need to calculate a 3×3 determinant,

$$\Delta = \sum_{j=1}^3 a_{1j} \text{cof}(a_{1j}) = \sum_{j=1}^3 a_{2j} \text{cof}(a_{2j}) = \sum_{j=1}^3 a_{3j} \text{cof}(a_{3j})$$

We can calculate determinant along any row or column of the matrix.

2.9.1 Properties of Determinants

- (i) If 'A' is a Square Matrix of size ' $n \times n$ ' and 'k' is a Scalar then

$$|K \cdot A_{n \times n}| = K^n \cdot |A_{n \times n}|$$
- (ii) $|adj(A)| = |A|^{(n-1)}$
- (iii) $|adj(adj(A))| = (|A|)^{(n-1)^2}$
- (iv) $|AB| = |A| \cdot |B|$
- (v) $|(AB)^T| = |B^T| \cdot |A^T|$
- (vi) If two rows (or) two columns of a determinant are interchanged, then the determinant changes its sign.
- (vii) The determinant of an upper triangular Matrix/a lower triangular Matrix/a diagonal Matrix is the product of the principal diagonal elements of the Matrix.
- (viii) The determinant of Every Skew-Symmetric Matrix of odd order ($A_{n \times n}$) ('n' is odd) is zero.
- (ix) The determinant of an orthogonal Matrix $A_{n \times n}$ is ± 1
- (x) The determinant of an Idempotent Matrix is either 0 (or) 1.
- (xi) The determinant of an Involutory Matrix is ± 1
- (xii) The determinant of a Nilpotent Matrix is always zero.
- (xiii) If the product of two Non-zero Matrices $A_{n \times n} \neq 0$; $B_{n \times n} \neq 0$ is a zero Matrix ($(AB)_{n \times n} = 0$), then both $|A| = 0$ & $|B| = 0$.
- (xiv) If two rows (or) two columns of a Matrix are either equal or Proportional, then the determinant of the Matrix is equal to zero.
- (xv) The number of terms in the general expansion of an ' $n \times n$ ' determinant is $n!$
- (xvi) Value of the determinant is invariant under row and column interchange i.e., $|A^T| = |A|$
- (xvii) If any row or column is completely zero, then $|A| = 0$.
- (xviii) If any single row or column of the matrix is multiplied by k then the determinant the of new matrix = $K|A|$
- (xix) In a determinant the sum of the product of the element of any row or column with its cofactor gives a determinant of the matrix.
- (xx) In determinant the sum of the product of the element of any row or column with a cofactor of another row or column will give zero.
- (xxi) $|AB| = |A| \times |B|$
- (xxii) Elementary operations don't effect the determinant that is $A \xrightarrow{R_i = R_i + KR_j} B$ then $|A| = |B|$
 $A \xrightarrow{C_i = C_i + KC_j} B$ then $|A| = |B|$

2.10 Minors, Cofactor and Adjoint of a Matrix

Minor of an element is equal to the determinant of the remaining elements of the matrix, after excluding the row and column containing the particular element. The cofactor of an element can be calculated from the minor of the element. The cofactor of an element is equal to the product of the minor of the element, and -1 to the power of position values of row and column of the element.

$$\text{Cofactor of an Element} = (-1)^{i+j} \times \text{Minor of an Element}$$

Here i and j are the positional values of the row and column of the element.

Example :

$$\text{If } \Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\text{Minor of element } a_{21}: M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\text{Co-factor of an element, } a_{ij} = (-1)^{i+j} M_{ij}$$

- To design co-factor matrix, we replace each element by its co-factor.
- Adjoint of a matrix = transpose of cofactor matrix
- $A^{-1} = \frac{\text{Adj}(A)}{|A|}$

2.11 Inverse of a matrix

Inverse of a matrix only exists for square matrices.

$$(A^{-1}) = \frac{\text{Adj}(A)}{|A|} \text{ and } |A| \neq 0$$

Properties:

- $AA^{-1} = A^{-1}A = I$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$
- $(A^T)^{-1} = (A^{-1})^T$
- The inverse of 2×2 matrix should be remembered,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{(ad-bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- Interchange the diagonal elements and put negative sign on the rest.
- Divide by determinant.